From Ratings to Returns: Can Peer Information Increase Agricultural Technology Adoption?

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Abstract

Many smallholder farmers in low-income countries do not adopt agricultural inputs sold by agro-dealers due to uncertainty about their performance. We introduce a system that aggregates farmers' evaluations of seed into ratings and disseminates them to farmers and agro-dealers. We test this approach in a randomized controlled trial in the market for high-yielding maize seed in Uganda. The intervention increases farmers' perceptions of seed performance, adoption, and yields. It draws farmers into the formal input market and raises productivity without inducing farmers to switch sellers or inducing agro-dealers to improve performance, suggesting that updating farmers' beliefs and reducing their uncertainty about seed performance are central to the effects of the rating system.

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1 Introduction

Adoption of modern agricultural inputs among smallholder farmers in low- and middle-income countries remains low, despite their potential to raise productivity and reduce poverty. As reviewed by Suri and Udry (2022), explanations include credit, liquidity, and savings constraints; limited access to insurance; information frictions; high transaction costs and poor infrastructure; and imperfections in labor and land markets.

A key information friction arises because farmers lack credible information about input performance, as it is unobservable prior to purchase and difficult to attribute even after use. Learning from own experience is challenging because agricultural outcomes depend on multiple factors beyond input performance (Bulte et al., 2025; Laajaj and Macours, 2026). At the same time, information about input performance is dispersed across farmers. Although they may exchange experiences within their social networks, such information often reflects a small number of peers and may be incomplete, selectively transmitted, or idiosyncratic.

These learning constraints allow uncertainty about input performance to persist, enabling the continued sale of poorly performing products at agro-dealers, the main distribution channel for agricultural inputs (Dar et al., 2024).

Reducing this uncertainty requires credible information about seed performance prior to purchase, yet the signals currently available to farmers are incomplete. Seed companies signal quality through labeling and packaging, but farmers struggle to identify low-quality seed without training (Gharib et al., 2021). Regulatory certification systems are costly, often constrained by institutional capacity, and do not always reliably signal seed performance (Tyack et al., 2026). Moreover, these signals only reach farmers who already engage with agro-dealers, and even certified seed may perform poorly when it is handled inadequately (Barriga and Fiala, 2020).

In this paper, we test an alternative system that collects farmers' evaluations of maize seed purchased from agro-dealers and aggregates them into ratings that summarize the experiences of multiple peers facing similar production conditions. These ratings are disseminated to both farmers and agro-dealers, providing a summary signal of seed performance at agro-dealers that individual farmers may find difficult to obtain through their own experimentation or social networks.

The rating system addresses key limitations of existing approaches. Ratings are disseminated directly to farmers, allowing information to reach those who do not typically visit agro-dealers. The system is decentralized and relies less on institutional capacity. Importantly, it does not directly measure seed purity or certify authenticity. Instead, it reflects farmers' evaluations of realized performance – whether the seed they purchased yielded well, germinated and matured reliably, and tolerated local conditions. While these assessments may diverge from laboratory measures of seed purity or authenticity, they capture performance dimensions that are relevant to farmers' production decisions.

Aggregating and disseminating farmers' purchasing experiences could improve outcomes through several channels. First, farmers may hold imprecise or biased beliefs about seed performance. Peer ratings could reduce uncertainty and increase confidence and, when prior beliefs are pessimistic relative to the signal, raise expected seed performance, thereby increasing adoption.

Second, agro-dealer-specific ratings could enable sorting. In agricultural input markets, farmers may choose among sellers with limited information about their relative seed performance. By revealing performance differences across agro-dealers, ratings could enable farmers to reallocate purchases toward higher-performing sellers (Klein, Lambertz, and Stahl, 2016; Jensen and Miller, 2018).

Third, the rating system could improve seed performance through reputational discipline. Agro-dealers may have weak incentives to invest in seed handling because farmers cannot assess seed performance at the point of purchase and struggle to attribute later outcomes, such as yield, to agro-dealer behavior. By aggregating farmers' experiences into visible ratings, the system could make selling poorly preserved seed reputationally costly, while rewarding better preserved seed with future demand, therefore shifting competition toward quality (Jin and Leslie, 2003; Hasanain, Khan, and Rezaee, 2023; Annan, 2025). These three channels are not mutually exclusive, but they generate distinct empirical predictions that allow us to assess their relative importance.

We test this approach in a randomized controlled trial (RCT) involving 3,470 smallholder farmers and 348 agro-dealers in the market for high-yielding maize seed in Uganda, where several studies document substantial performance uncertainty (Bold et al., 2017; Barriga and Fiala, 2020; Ilukor et al., 2025).[1] The randomized design allows us to estimate causal effects, while three rounds of detailed data enable us to understand why the intervention affects behavior. We track outcomes along the full causal chain, from farmers' perceptions and adoption decisions to agricultural productivity and sales. Additional data on farmers' choice of agro-dealers, agro-dealer seed handling practices, services, and signaling, and objectively measured seed-performance indicators allow us to trace how the intervention operates.

Exposure to aggregated peer ratings increases adoption of high-yielding maize seed and purchases from agro-dealers, and treated farmers achieve maize yields that are 50 kg/acre higher than those of the control group at endline, equivalent to 11% of the baseline mean. The value of maize production net of seed costs is also 66,000 UGX higher for treated farmers. Because adoption and productivity effects are consistent with all three channels, interpretation rests on outcomes that are predicted to differ across pathways. The evidence points most strongly to belief updating: the intervention raises farmers' perceptions of agro-dealer seed performance, while switching across agro-dealers increases modestly at midline but is not persistent. On the supply side, agro-dealers respond along service and signaling margins, but we do not detect improvements in seed handling, storage, or measured seed performance.

This paper contributes to the literature on agricultural technology adoption under input-quality uncertainty, misperceptions, and imperfect learning (Wossen et al., 2022; Michelson et al., 2023; Wossen et al., 2024; Bohr et al., 2024). We advance this literature

[1] In this study, the term "high-yielding maize varieties" refers to both open-pollinated varieties (OPVs) and hybrids that have been genetically improved through breeding programs. We explain the terminology and our rationale for using this terminology in Appendix A.1.

in three ways. First, we introduce an intervention that addresses limitations of existing seed performance signals – it is decentralized, experience-based, and reaches farmers who do not engage with formal markets – and show that it improves both adoption and productivity. Second, the pattern of results – positive treatment effects on perceptions, adoption, and productivity, alongside limited support for switching and seed performance improvements – provides evidence that uncertainty about seed performance is a binding constraint on adoption. While farmers' imprecise or overly pessimistic beliefs have been documented in fertilizer markets (Michelson et al., 2021; Hoel et al., 2024), we extend this evidence to seed markets. Third, by tracking outcomes on both sides of the market – farmers' beliefs and behavior, as well as agro-dealers' responses and seed performance – we assess whether belief updating, agro-dealer sorting, or reputational discipline is most consistent with the pattern of results. Together with Dar et al. (2024), our study is among the few to link farmer and agro-dealer perspectives, providing a more complete picture than studies focusing on either side in isolation.

More broadly, this paper relates to a growing literature on public quality information in markets with hard-to-observe seller performance. Research on online review platforms documents that consumer feedback affects both sides of the market: buyers use crowd-sourced reviews to make more informed choices, while sellers adjust their behavior in response to publicly observable quality signals (Dranove and Jin, 2010; Klein, Lambertz, and Stahl, 2016; Reimers and Waldfogel, 2021). A related literature shows that reducing search and information frictions can improve matching, trade, and market functioning in both online and low-income-country markets (Jensen and Miller, 2018; Bergquist, McIntosh, and Startz, forthcoming; Startz, forthcoming; Dillon, Aker, and Blumenstock, 2025). In several of these settings, key channels are sorting – information helps buyers identify better sellers – and reputational discipline, whereby information shifts competition toward quality.

Our study shows that in an offline agricultural input market characterized by uncertainty about input performance, crowd-sourced ratings need not operate primarily through seller sorting or quality competition. Instead, the evidence points predominantly to belief updating about agro-dealer seed performance. The limited support for switching is consistent with Hasanain, Khan, and Rezaee (2023), who find that crowd-sourced information on artificial livestock insemination services in Pakistan does not induce farmers to switch providers. The limited support for quality upgrading is consistent with Hsu and Wambugu (2024), who show that training farmers in rural Kenya to identify quality-verified seeds does not lead to quality improvements by local sellers.

Finally, the paper connects to the literature on peer learning in agricultural technology adoption (Foster and Rosenzweig, 1995; Bandiera and Rasul, 2006; Conley and Udry, 2010), but differs by aggregating dispersed individual experiences into a public rating system rather than studying learning through social networks.

The remainder of the paper is organized as follows. Section 2 presents the conceptual model, Section 3 describes the experimental design, Section 4 outlines the empirical strategy, Section 5 describes the data, Section 6 presents the results, and Section 7 concludes.

2 Conceptual model

This section develops a model of how aggregated peer information affects farmer and agro-dealer outcomes, generating a set of predictions that guide the empirical analysis.

2.1 Setup

We assume that smallholder farmers choose between purchasing high-yielding seed varieties from agro-dealers and relying on outside options such as seed they saved themselves or obtained through informal exchanges. Let θ_j denote the performance of seed sold by agro-dealer j relative to farmer i's outside option. There are J agro-dealers in farmer i's area, each with performance θ_j. This is the actual performance the farmer would realize if they planted the seed, unobservable at the time of purchase.

We decompose θ_j as $\theta_j = \theta_0 - d(e_j)$, where θ_0 is the intrinsic performance of the seed upon arrival at agro-dealer j, determined upstream in the supply chain, outside the agro-dealer's control, and assumed to be the same across agro-dealers,[2] and $d(e_j) \geq 0$ captures performance degradation due to handling and storage, with $d'(e_j) < 0$. Higher agro-dealer effort e_j (e.g., proper storage containers, ventilation, temperature, and moisture control) reduces degradation, preserving seed performance closer to its intrinsic level θ_0.

Because seed performance is unobservable at the time of purchase, the decision to buy seed depends on beliefs about the performance of agro-dealer seed relative to outside options $E_i[\theta_j]$. Seed has some properties of a credence good: because agricultural outcomes depend on many factors beyond the seed itself and individual learning is noisy, slow, and difficult to self-correct, farmers cannot fully assess seed performance even after use (Ahmad, 2022; Laajaj and Macours, 2026). This is compounded by the disproportionate salience of negative information (Rozin and Royzman, 2001; Vaish, Grossmann, and Woodward, 2008; Hornik et al., 2015), which may push beliefs below the true θ. Evidence from fertilizer markets (Michelson et al., 2021; Hoel et al., 2024) and maize seed in Uganda (Barriga and Fiala, 2020) is consistent with potentially downward-biased beliefs. At the same time, seed also has some properties of an experience good: farmers who have purchased and planted agro-dealer seed can form more accurate beliefs than those who have not (Foster and Rosenzweig, 1995).

Farmers are risk-averse and face a two-part decision: whether to purchase seed from any agro-dealer rather than rely on an outside option (extensive margin), and if so, from which agro-dealer (agro-dealer choice). Farmer i evaluates each agro-dealer j by comparing the expected performance of that agro-dealer's seed, discounted for uncertainty, against the purchase cost. Formally, the farmer purchases from the best available agro-dealer if the certainty equivalent exceeds the cost:

[2] This assumption abstracts from upstream variation in seed received by agro-dealers; allowing θ_0 to vary by agro-dealer would introduce an additional source of agro-dealer heterogeneity but would not affect the model's distinction between belief updating, agro-dealer sorting, and agro-dealer reputational discipline.

$$\max_j \; E_i[\theta_j] - \frac{\rho}{2}\sigma_{ij}^2 - c_j > 0 \tag{1}$$

where σ_{ij}^2 is farmer i's subjective uncertainty about agro-dealer j's seed performance, $\rho > 0$ is the coefficient of absolute risk aversion, and $c_j > 0$ is the cost of purchasing from agro-dealer j relative to the outside option. Adoption depends on both expected performance and subjective performance uncertainty: even at the same expected return, greater uncertainty discourages adoption.

2.2 Belief updating

Farmer i does not observe θ_j; instead, they hold a prior belief $\mu_{ij} \equiv E_i[\theta_j]$ about agro-dealer j's seed performance: $\theta_j \mid \mathcal{I}_i \sim \mathcal{N}(\mu_{ij}, \sigma_{ij}^2)$. Priors are likely to be imprecise or biased. The intervention can affect outcomes by shaping belief formation about θ_j. In our model, the way it does so depends on farmers' prior seed purchasing experience from agro-dealers, generating predictions about heterogeneous treatment effects. In particular, we distinguish two types:

- **Purchasers** ($i \in P$) have prior beliefs informed by their own purchasing experience; as a result, μ_{Pj} is close to θ_j.
- **Non-purchasers** ($i \in NP$) lack own purchasing experience: we assume $\sigma_{NPj}^2 > \sigma_{Pj}^2$ (greater uncertainty), and allow μ_{NPj} to differ from θ_j. Farmers without experience hold more biased and less precise priors. In the empirically relevant case of pessimistic priors, $\mu_{NPj} < \theta_j$.

The intervention provides farmer i with a signal about the performance of seed from agro-dealer j. The signal aggregates the experiences of n_j farmers who purchased from agro-dealer j:

$$s_j = \frac{1}{n_j}\sum_{k=1}^{n_j}(\theta_j + \varepsilon_k) = \bar{\theta}_j + \bar{\varepsilon}_j \tag{2}$$

where ε_k is the noise in rater k's (drawn from the pool of previous purchasers) individual experience (reflecting weather, soil, management, and other factors beyond seed performance) and $\bar{\varepsilon}_j \sim \mathcal{N}(0, \sigma_\varepsilon^2/n_j)$.

To characterize how such a signal changes beliefs, we use the standard conjugate Normal-Normal Bayesian updating framework. Given farmer i's prior belief about agro-dealer j's seed performance μ_{ij} and the signal s_j, posterior mean and variance are:

$$\tilde{\mu}_{ij} = (1-\alpha_{ij})\mu_{ij} + \alpha_{ij}s_j, \quad \tilde{\sigma}_{ij}^2 = \frac{\sigma_{ij}^2 \cdot \sigma_\varepsilon^2/n_j}{\sigma_{ij}^2 + \sigma_\varepsilon^2/n_j} \tag{3}$$

with the weight on the signal, $\alpha_{ij} = \frac{\sigma_{ij}^2}{\sigma_{ij}^2 + \sigma_\varepsilon^2/n_j}$, increasing in prior subjective uncertainty σ_{ij}^2.[3] Since $\sigma_{NPj}^2 > \sigma_{Pj}^2$, non-purchasers place more weight on the signal and experience larger reductions in posterior uncertainty. The belief shift $\tilde{\mu}_{ij} - \mu_{ij} = \alpha_{ij}(s_j - \mu_{ij})$ is positive when the signal is more favorable than the prior. For purchasers, the main effect is a precision gain (lower $\tilde{\sigma}_{ij}^2$); for non-purchasers, the model predicts a larger precision gain and, when priors are pessimistic relative to the signal, a larger upward revision in expected performance.

We interpret s_j as an informative peer-experience signal rather than an unbiased estimate of agro-dealer-specific performance, since it need not represent the experience that any farmer would have with agro-dealer j. Raters are drawn from previous purchasers and may differ from non-purchasers along dimensions beyond prior beliefs, but the model requires that average rater experiences contain information about agro-dealer seed performance beyond selection-driven differences across farmers. Under this condition, averaging across raters reduces idiosyncratic noise: more raters generate a more precise signal about agro-dealer j's seed performance.

2.3 Agro-dealer sorting

While the belief-updating channel captures changes in farmers' beliefs about agro-dealer seed relative to outside options, agro-dealer sorting captures changes in seller choice conditional on purchasing from an agro-dealer. Because the intervention provides agro-dealer-specific ratings (s_j), it conveys two types of information. First, the average signal across agro-dealers, $\bar{s} = \frac{1}{J}\sum_j s_j$, provides information about overall seed performance, allowing farmers to update beliefs about agro-dealers as a group. Second, differences in signals across agro-dealers and deviations from the average, $s_j - \bar{s}$, reveal performance differences and enable sorting.

Whether sorting occurs depends on the dispersion in θ_j across agro-dealers. If performance varies substantially across agro-dealers and ratings reveal this, farmers can switch to higher-rated agro-dealers. If ratings are similar across agro-dealers, the signal contains little agro-dealer-specific information, weakening the sorting channel.

2.4 Reputational discipline

The intervention can also induce agro-dealers to reduce degradation $d(e_j)$, thereby increasing performance θ_j itself. This operates through a reputational discipline channel: by aggregating farmers' experiences into an observable signal, the intervention raises the reputational cost of exerting low effort and selling poorly preserved seed. Once ratings are disseminated to farmers, agro-dealers' performance becomes visible to potential buyers.

[3] The Bayesian updating expression should be interpreted as a benchmark in which farmers perceive the rating signal as credible and relevant. Lower credibility or attention would attenuate the weight placed on s_j, without altering the model's qualitative predictions.

Let $e_j \geq 0$ denote agro-dealer j's effort in seed preservation (e.g., proper storage containers, ventilation, temperature, and moisture control), with convex cost $c(e_j)$. Recall that $\theta_j = \theta_0 - d(e_j)$ with $d'(e_j) < 0$: higher effort in seed storage and handling reduces degradation, preserving seed performance closer to its intrinsic level θ_0. Without the intervention, agro-dealers' reputational incentives are limited to repeat purchases and informal word of mouth. Let $R_C(\theta_j)$ denote demand for agro-dealer j in the absence of the intervention, that depends on quality through these informal channels, with $R_C' > 0$. Under the intervention, ratings are collected and disseminated, strengthening the reputational return to quality. Let $R_T(\theta_j)$ denote demand under the rating system, with $R_T'(\theta_j) > R_C'(\theta_j)$: the marginal return to quality is higher when performance is publicly visible. The agro-dealer chooses effort to maximize:

$$\max_{e_j} \quad R_\tau(\theta_0 - d(e_j)) - c(e_j), \quad \tau \in \{C, T\} \tag{4}$$

The first-order condition is $R_\tau'(\theta_0 - d(e_j^*))(-d'(e_j^*)) = c'(e_j^*)$. Since $R_T' > R_C'$, the optimal effort is higher under the intervention: $e_{jT}^* > e_{jC}^*$. If agro-dealers respond to the intervention by investing in seed preservation, degradation falls and performance improves. If the belief-updating and reputational discipline channels operate simultaneously, they may reinforce each other: updated beliefs increase farmers' willingness to engage with agro-dealers, while better seed preservation strengthens the actual purchasing experience.

2.5 Testable predictions

The model identifies three channels through which the intervention may affect outcomes: belief updating (Subsection 2.2), agro-dealer sorting (Subsection 2.3), and reputational discipline (Subsection 2.4). Each channel generates predictions across five outcomes, summarized in Table 1.

Prediction 1 (Perceptions). Under the belief-updating channel, the treatment shifts perceived seed performance in the direction of the signal, with stronger effects among non-purchasers. If priors are pessimistic relative to the signal, perceptions shift upward. Under the agro-dealer sorting channel, there may be an indirect effect on perceptions if farmers sort to higher-rated agro-dealers and subsequently experience better seed performance. Under the reputational-discipline channel, perceptions may also improve indirectly if agro-dealers adjust their effort and improved objective seed performance becomes visible. For both channels, effects need not be concentrated among non-purchasers.

Prediction 2 (Adoption). Under the belief-updating channel, the treatment increases adoption by reducing uncertainty and, if priors are pessimistic relative to the signal, by raising expected performance above the adoption threshold. Under the agro-dealer sorting and reputational-discipline channels, the treatment may increase adoption through indirect effects on farmers' perceptions.

Prediction 3 (Productivity). All three channels can increase yields if they increase adoption of genuinely higher-performing seed relative to outside options. Agro-dealer sorting and reputational discipline may additionally increase yields conditional on adoption – through sorting toward better agro-dealers and through improving objective seed performance, respectively – thereby enhancing the performance of seed actually purchased and planted.

Prediction 4 (Switching). Under the belief-updating channel, the treatment does not reallocate farmers across agro-dealers. Under the agro-dealer sorting channel, the treatment increases switching toward higher-rated agro-dealers, as agro-dealer-specific ratings reveal performance differences. Absence of persistent increases in switching, or switching not directed toward higher-rated agro-dealers, weakens the evidence for agro-dealer sorting as the primary channel. Under the reputational-discipline channel, the treatment does not affect switching.

Prediction 5 (Seed performance). Under the belief-updating channel, the treatment shifts perceived but not objective seed performance at a given agro-dealer. Under the agro-dealer sorting channel, the treatment changes the performance of seed actually purchased at the farmer level when farmers switch toward better agro-dealers, but this does not operate through product improvement at the agro-dealer level. Under the reputational-discipline channel, seed degradation falls and performance improves as agro-dealers respond to reputational incentives by increasing effort in seed preservation. Finding no treatment effect on agro-dealer-level seed performance weakens the evidence for the reputational-discipline channel.

The channels may also differ in their predicted timing of effects. Belief updating may shift perceptions as soon as farmers receive ratings, with effects on adoption materializing at the next purchase occasion. Agro-dealer sorting may induce switching at the next purchase occasion, while effects on perceptions and adoption may materialize later, when farmers experience better seed performance at higher-rated agro-dealers. Reputational discipline may first induce agro-dealer improvements in seed storage and handling, which then translate into improved seed performance and subsequently become observable to farmers through experience or updated ratings; effects on perceptions and adoption may therefore take longer to materialize. This timing distinction is useful for interpreting whether effects appear at midline, persist through endline, or emerge only after an additional rating cycle.

3 Experimental design

3.1 Randomization and power

Randomization is conducted at the agro-dealer catchment-area level with equal assignment probability. Catchment areas consist of geographically proximate clusters of towns, villages, markets, trading centers, and other key hubs, and typically include sev-

Table 1: Model predictions by channel and outcome

	Belief updating	**Agro-dealer sorting**	**Reputational discipline**
Perceptions	Shift directly toward signal, larger for non-purchasers	Shift indirectly, uniform across farmers	Shift indirectly, uniform across farmers
Adoption	Increases via lower uncertainty or higher expected performance	Increases via perceptions	Increases via perceptions
Productivity	Increases via adoption	Increases via adoption & sorting toward better dealers	Increases via adoption & improved seed performance
Switching	No effect	Increases toward higher-rated dealers	No effect
Seed performance	No effect	No effect	Improves via increased dealer effort

Note: Heterogeneity in perceptions (P1), limited switching (P4), and no agro-dealer-level improvements in seed performance (P5) would jointly point to belief updating as the dominant channel.

eral agro-dealers.[4] Treatment is assigned at the catchment-area level rather than within areas (e.g., at the agro-dealer level), ensuring that all farmers within a catchment area are exposed to agro-dealers under the same treatment condition, which enables identification of farmer-level treatment effects.

Sample sizes were determined using simulation-based power calculations based on data from prior surveys, allowing power to be assessed under empirically relevant outcome distributions at both the farmer and agro-dealer levels. Full details are provided in the pre-analysis plan (AEA RCT Registry, RCT ID 0006361).

3.2 Intervention

Rating collection and computation

Treated farmers were asked to rate maize seed sold by agro-dealers in their catchment area. Farmers could rate an agro-dealer only if they had personally purchased seed from that agro-dealer or knew someone who had. Enumerators guided farmers through

[4] To define these areas, we used a haversine function to construct an adjacency matrix based on GPS coordinates, assigning agro-dealers less than five kilometers apart to the same catchment area. The five-kilometer threshold was selected based on a visual inspection of the map, the average village size in our sample, and the reported distance between farmers and agro-dealers in survey data from a previous study of the maize value chain, available here.

Table 2: Questions for farmers to rate agro-dealers

	min	max
Do you know this *shop name* or *dealer name*, sometimes called *nickname*, located in *market name*? The place can be described as *description*.	no	yes
Please rate this agro-dealer on:		
Quality and authenticity of seed	1 star	5 stars
Please rate the maize seed that this agro-dealer sells on:		
General quality	1 star	5 stars
Yield as advertised	1 star	5 stars
Drought tolerance as advertised	1 star	5 stars
Pest/disease tolerance as advertised	1 star	5 stars
Speed of maturing as advertised	1 star	5 stars
Germination	1 star	5 stars

the list of agro-dealers using a tablet-based application displaying names, location descriptions, and storefront pictures collected at the start of the study (see Subsection 5.1).

Ratings were collected post-harvest, when farmers could form noisy assessments of seed performance based on realized outcomes such as germination, yield, drought resilience, pest and disease resistance, and crop maturation speed, using the questions listed in Table 2 and a five-point star scale. The rating dimensions capture attributes that farmers attend to when purchasing and using seed, as identified through prior qualitative work, including workshops with experts from MAAIF, NARO, USTA, UNADA, Naseco Seed Company, and extension specialists.

Because agro-dealers stock heterogeneous seed varieties with different agronomic profiles, assessing overall seed performance in absolute terms may obscure trade-offs across characteristics. To address this, the rating instrument was designed to capture performance relative to product claims. Farmers were asked to rate attributes "as advertised," e.g. by evaluating "maize seed sold by the agro-dealer" on "yield as advertised." This approach allows performance to be assessed on a comparable basis across varieties while preserving an agro-dealer-level performance signal: an agro-dealer selling a variety without stress-tolerance traits that performs as advertised can receive a high rating, whereas an agro-dealer selling a stress-tolerant variety that underperforms relative to its claims receives a lower rating.

A potential concern is that the rating task itself could increase farmers' awareness of agro-dealers, potentially confounding treatment effects. To address this, control-group farmers were also guided through the list of agro-dealers in their catchment area in a similar manner (displaying names, location descriptions, and storefront pictures), ensuring comparable exposure. However, they were not asked to provide ratings, as the act of rating is a component of the treatment. This design also reduces the scope for reactivity and experimenter bias.

Average ratings for each agro-dealer were computed by aggregating all farmer responses within the catchment area. Details on the rating computation are provided in Appendix A.2.

Rating dissemination to farmers

The treatment relied on timely dissemination of agro-dealer ratings before the next agricultural season, allowing farmers to incorporate this information into seed purchasing decisions. Dissemination occurred through short message service (SMS) and in-person visits.

Farmers received one SMS per agro-dealer in their catchment area, delivered in one of three local languages – Lusoga, Lugwere, or Samia. For farmers in treatment areas, the message read:

> Hello from AgroAdvisor! Did you know that customers rate the quality of maize seed sold at [*name of the agro-dealer*] as [*okay/good/very good/excellent*]?

Control farmers received a placebo message that merely highlighted the existence of agro-dealers in their catchment area.

In addition to SMS dissemination, enumerators revisited farmers and displayed agro-dealer ratings using a tablet-based application. The application cycled through all agro-dealers in the catchment area (again displaying names, location descriptions, and storefront pictures), presenting this message accompanied by a one-to-five-star visual rating:

> We wanted to let you know that customers rate the quality of maize seed sold at [*name of the agro-dealer*] as [*okay/good/very good/excellent*]! The maize seed from this agro-dealer received a score of [*score*] out of 5!

For control group farmers, the application similarly cycled through agro-dealers in the catchment area without displaying ratings. Together with the placebo SMS, this design isolates the effects of rating information from potential reactivity or salience effects arising from reminding farmers of the existence of agro-dealers in their area. More details on the rating dissemination are provided in Appendix A.2.

Rating dissemination to agro-dealers

Agro-dealers received their ratings in the form of a report. The front page displayed a certificate with a logo and the agro-dealer's overall rating. The back page provided more detailed information, presenting ratings for seed sold by the agro-dealer across the six key dimensions and a table displaying average ratings of other agro-dealers in the same catchment area. An example of the full report is provided in Appendix A.3.

The study spans two agricultural seasons (the second season of 2021 and the first season of 2022) with three rounds of data collection. Before season 1, we collect ratings from treated farmers, aggregate them, and disseminate them to farmers and agro-dealers prior to planting. Between seasons 1 and 2, we collect a second round of ratings and disseminate the updated ratings prior to planting in season 2. After season 2, we collect final ratings as an outcome measure, but these are not disseminated.

4 Empirical strategy

We estimate intention-to-treat effects using Ordinary Least Squares:

$$Y_{ij} = \alpha + \beta T_j + \gamma' X_{ij} + \delta Y_{0ij} + \varepsilon_{ij} \tag{5}$$

where Y_{ij} is the outcome for farmer i in catchment area j, Y_{0ij} is the corresponding mean-centered baseline outcome, T_j is a binary indicator for treatment assignment, and X_{ij} includes controls for orthogonal treatments in the factorial design, as the study was embedded in a larger project with additional cross-randomized interventions (demeaned and interacted with the main treatment effect; see Lin, 2013; Muralidharan, Romero, and Wüthrich, 2023). The coefficient β captures the intention-to-treat effect. We estimate analogous specifications for agro-dealer-level outcomes, where Y_{ij} denotes the outcome for agro-dealer i in catchment area j.

Since randomization occurs at the catchment area level, we cluster standard errors accordingly. For farmer-level outcomes, we use the conventional sandwich estimator without small-sample corrections. For agro-dealer-level outcomes, we apply the Bell-McCaffrey adjustment to account for the smaller number of observations (Imbens and Kolesár, 2016).

Following our pre-analysis plan, we trim continuous outcomes to limit the influence of outliers and apply inverse hyperbolic sine transformations to skewed variables. Outcomes with limited variation, defined as having 95% or more of observations at a single value within the relevant sample, are excluded from the analysis.

To account for multiple hypothesis testing, we aggregate related outcomes into summary indices following Anderson (2008), which assigns lower weights to highly correlated outcomes and thereby concentrates information.[5] As a robustness check, we also compute summary indices following Kling, Liebman, and Katz (2007) in Appendix A.4. While these indices capture overall effects across outcome families, we also report treatment effects on individual outcomes to help interpretation, with individual significance treated cautiously.

[5] In regressions using these overall indices, we do not control for baseline values, as doing so would restrict the sample to units with complete data for all constituent variables at baseline and the respective follow-up wave (midline or endline), thereby substantially reducing statistical power.

5 Data

5.1 Sample

The agro-dealer sample was constructed by listing all agro-dealers across eleven districts in southeastern Uganda, drawing on official records of registered agro-dealers and extensive field visits to identify new and informal ones. 348 eligible agro-dealers were assigned to 130 catchment areas (see Subsection 3.1 for details). Of the 130 catchment areas, 65 were assigned to treatment, covering 193 agro-dealers, and 65 to control, covering 155 agro-dealers. Across all catchment areas, the average number of agro-dealers per area was three, with the number ranging from one to 18.

To link agro-dealers with farmers, agro-dealers identified the villages where most of their customers reside. Enumerators then randomly sampled ten maize-growing households from these villages, yielding a sample of 3,470 smallholder farmers.

Baseline data were collected from agro-dealers in September-October 2020 and from farmers in April 2021; midline data were collected in January-February 2022, and endline data in July-August 2022.

At the agro-dealer level, enumerators interviewed the individual most knowledgeable about day-to-day operations, purchased a maize seed bag, and recorded indicators related to seed storage conditions. At the farmer level, enumerators interviewed the individual most knowledgeable about maize farming. In addition to general questions, farmers listed all maize plots, from which one was randomly selected for detailed data collection, thereby avoiding selective reporting across plots and allowing us to collect plot-level data on adoption and harvest outcomes.

5.2 Baseline descriptives and randomization balance

Table 3 reports farmer characteristics at baseline. The average household head is 49 years old, 78% are male, and 51% have completed primary education. Households have nine members on average and are located four kilometers from the nearest agro-dealer and nine kilometers from the nearest tarmac road. Farmers cultivate three acres, have 23 years of maize-growing experience, and reported yields of 440 kilograms per acre in the season preceding the baseline survey.

Half of farmers planted high-yielding maize seed on at least one plot in that season.[6] One in three purchased it from an agro-dealer. Other farmers obtained high-yielding seed from Operation Wealth Creation/National Agricultural Advisory Services or non-governmental organizations; many relied on their own retained seed or farmer-to-farmer exchanges. These patterns document that farmers have meaningful outside options to purchasing seed from agro-dealers. At the same time, 68% of farmers report believ-

[6]"High-yielding maize varieties" refers to both open-pollinated varieties and hybrids that have been improved through breeding programs. "Farmer-saved seed" refers to seed of unknown progeny or quality, which typically performs worse than freshly purchased high-yielding maize varieties from agro-dealers. Our terminology and the rationale for these definitions are detailed in Appendix A.1.

ing that maize seed sold by agro-dealers is adulterated. This widespread skepticism underscores that farmers perceive agro-dealer seed as risky, implying that credible information about seed performance could affect adoption decisions by altering either expected performance or uncertainty.

Table 4 reports baseline characteristics of agro-dealers and their shops at baseline. The average respondent is 32 years old, 60% are male, and over 90% have completed primary education. In 55% of cases, the respondent is also the owner of the agro-shop. On average, agro-shops were established five years before the baseline survey, are located seven kilometers from the nearest tarmac road, and serve 41 customers per day. 74% sell only farm inputs and equipment, 60% provide credit, and 46% offer advisory services.

With the shop manager's permission, enumerators inspected seed storage areas and recorded observable characteristics. They reported pest issues (e.g., rats or insects) at 65% of agro-shops and maize seed stored in open containers at 16% of agro-shops. Enumerators also sampled maize seed bags at the agro-shops and measured their moisture content. Moisture is a commonly used objective indicator of seed performance because it directly affects germination and viability; additional details are provided in Appendix A.1. The average moisture content was 13.6%. Regarding packaging and labeling, 68% of purchased seed bags displayed a packaging date, 18% an expiry date, and only 8% a quality certification label issued by the National Seed Certification Services (NSCS), consistent with weak baseline guarantees of seed performance provided through agro-shops.[7]

Tables 3 and 4 also double as balance tables. In particular, we present standard orthogonality tests with pre-registered variables to assess baseline comparability between treatment and control groups. For farmer-level outcomes (Table 3), no comparison is significant; for agro-dealer-level outcomes (Table 4), two of 17 comparisons are significant at the 10% level. Joint F-tests further confirm baseline balance.

5.3 Attrition

Table 5 reports attrition rates for treatment and comparison groups. At midline, 12% of agro-dealers and 2% of farmers were missing; at endline, 14% of agro-dealers and 1% of farmers were not surveyed. To assess whether attrition was linked to treatment, we regressed the likelihood of leaving the sample on the treatment indicator and found that treated agro-dealers were significantly less likely to drop out.[8]

Differential attrition was further assessed by regressing pre-registered balance variables (see Table 4) on indicators for leaving the sample, for belonging to the control group, and on their interaction (Appendix A.5). Results indicate that attrition among

[7] In Uganda, maize seed certification is overseen by the NSCS within MAAIF. Certified seed is identified by a blue tag or sticker.

[8] Reasons for agro-dealer attrition at endline were: eleven shops closed, ten relocated, eight changed product offerings, three merged with other shops, one refused to be interviewed, and 17 reported other (4) or no specific reasons (13).

Table 3: Descriptive statistics and orthogonality tests - Farmer

	mean	treated	obs.
Household head's age in years†	48.62	-0.24	3453
	(13.38)	(0.56)	
Household head is male†	0.78	0.03	3470
	(0.42)	(0.03)	
Household head finished primary education†	0.51	0.04	3437
	(0.50)	(0.03)	
Homestead's distance to nearest tarmac road in km	9.39	-1.23	3302
	(10.81)	(1.71)	
Homestead's distance to nearest agro-dealer in km†	3.78	0.11	3339
	(4.79)	(0.37)	
Number of people in household (incl. respondent)†	8.70	-0.09	3470
	(3.98)	(0.18)	
Number of rooms in house	3.49	0.02	3469
	(1.45)	(0.09)	
Farmer's land for crop production in acres	3.35	0.00	3442
	(4.32)	(0.22)	
Years since farmer started growing maize	23.09	-0.55	3470
	(13.14)	(0.58)	
Yield in kg/acre†	442.06	-3.99	3279
	(303.91)	(13.66)	
Farmer used high-yielding maize varieties on any plot†	0.49	0.01	3466
	(0.50)	(0.02)	
Farmer bought this seed at agro-dealer†	0.32	0.01	3424
	(0.47)	(0.02)	
Amount of this seed farmer bought at agro-dealer in kg†	9.52	-0.34	1064
	(6.92)	(0.53)	
Farmer thinks maize seed at agro-dealer is adulterated†	0.68	0.00	2673
	(0.46)	(0.03)	
Farmer used DAP/NPK	0.25	0.02	3465
	(0.43)	(0.04)	
Farmer used organic manure	0.07	0.01	3466
	(0.26)	(0.01)	

Note: Column (1) reports sample means at baseline and standard deviations below; column (2) reports differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; column (3) reports the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels. † indicates that the variable was pre-registered to test balance and is included in the F-test.

Table 4: Descriptive statistics and orthogonality tests - Agro-dealer

	mean	treated	obs.
Respondent's age in years†	32.43	-2.24+	347
	(11.49)	(1.21)	
Respondent is male†	0.59	-0.01	348
	(0.49)	(0.06)	
Respondent finished primary education†	0.92	-0.01	339
	(0.27)	(0.03)	
Respondent owns shop	0.55	0.02	348
	(0.50)	(0.06)	
Respondent received training on maize seed handling†	0.53	0.12+	348
	(0.50)	(0.07)	
Respondent knows how to store seed after repackaging†	0.27	0.08	348
	(0.44)	(0.06)	
Agro-dealer's distance to nearest tarmac road in km†	6.56	-1.58	343
	(10.39)	(2.24)	
Agro-dealer only sells farm inputs	0.74	0.03	348
	(0.44)	(0.06)	
Years since agro-dealer establishment†	5.34	0.21	348
	(6.30)	(0.78)	
Number of customers per day†	41.49	6.43	346
	(46.49)	(6.72)	
Quantity of maize seed sold in kg†	695.50	176.31	340
	(1497.18)	(235.92)	
Amount of maize seed lost/wasted last season in kg†	3.50	2.40	348
	(18.65)	(2.30)	
Number of maize varieties in stock	2.83	0.19	344
	(1.59)	(0.21)	
Agro-dealer has problem with pests	0.65	-0.03	348
	(0.48)	(0.06)	
Agro-dealer stores maize seed in open containers	0.16	0.08	348
	(0.36)	(0.05)	
Agro-dealer received seed related complaint from customer	0.64	0.07	348
	(0.48)	(0.05)	
Moisture in bag of maize seed in %	13.56	-0.18	226
	(1.44)	(0.26)	

Note: Column (1) reports sample means at baseline and standard deviations below; column (2) reports differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; column (3) reports the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels. † indicates that the variable was pre-registered to test balance and is included in the F-test.

Table 5: Attrition

	midline			*endline*		
	mean	treated	obs.	mean	treated	obs.
Agro-dealer left the sample	0.121	-0.108**	348	0.144	-0.079+	348
	(0.326)	(0.035)		(0.351)	(0.042)	
Farmer left the sample	0.018	0.001	3470	0.008	-0.001	3470
	(0.134)	(0.005)		(0.091)	(0.003)	

Note: Columns (1) and (4) report sample means at mid- and endline and standard deviations below; columns (2) and (5) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (6) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels.

agro-dealers is unlikely to bias our estimates. No differential attrition was observed at the farmer level; since primary outcomes are measured among farmers, attrition is unlikely to meaningfully affect the main treatment effects.

5.4 Rating data

At endline, the rating intervention generated a dyadic dataset comprising of 16,823 farmer-dealer links. Among farmers who provided at least one rating, the average farmer rated 1.4 agro-dealers (min = 1, max = 10). The average agro-dealer was rated by 6.7 farmers (min = 0, max = 30).

Because ratings could be provided either by farmers who had personally purchased seed from the agro-dealer or by those who knew someone who had, the signal may reflect indirect information rather than firsthand performance experience. In practice, 93% of ratings were provided by farmers with direct purchase experience, and ratings are similar between purchasers and non-purchasers (see Appendix A.6 for details). This suggests that the aggregate signal predominantly reflects direct experience.

Ratings may also reflect factors unrelated to seed performance, such as customer service or personal relationships. Appendix A.6 provides suggestive evidence that ratings are associated with observable characteristics plausibly related to seller and seed performance: specialized agro-shops and dealers with better seed handling practices receive higher ratings, and seed ratings on randomly selected fields are positively correlated with maize yields. These patterns are consistent with ratings reflecting farmers' assessments of seed performance rather than purely non-performance attributes. Appendix A.6 provides additional detail on the rating data, including the distribution of ratings across the six performance dimensions and their inter-correlations.

6 Results

We report treatment effects for both farmers and agro-dealers at midline (after one agricultural season) and at endline (after two seasons). All outcome variables were pre-specified in a pre-analysis plan at the AEA's registry for RCTs. The econometric analysis was run on simulated data prior to outcome data collection and documented in a registered report (Humphreys, De la Sierra, and Van der Windt, 2013). All documents, code, and data are publicly available in a version-controlled GitHub repository.[9]

Our primary approach to interpreting mechanisms relies on the pattern of effects across outcomes: each channel generates a distinct empirical signature across perceptions, adoption, productivity, switching, and seed performance (Table 1). We also examine heterogeneity by prior seed purchasing experience. Although this analysis was not pre-specified and is therefore exploratory, it is motivated by the model's prediction that farmers with greater uncertainty about seed performance should place more weight on the signal.

6.1 Perceived seed performance

We begin by testing whether the treatment shifts farmers' perceptions of maize seed performance at agro-dealers. Table 6 reports treatment effects on a perception index combining farmer ratings across six dimensions: general quality, yield, drought tolerance, pest and disease tolerance, maturity time, and germination. The treatment increases this index by 0.09 standard deviations, significant at the 10% level. The positive effect is consistent with the belief-updating channel if the peer signal is more favorable than farmers' prior beliefs about agro-dealer seed performance.

These estimates should be interpreted with caution. Ratings were collected from control farmers only at endline, and the perception index requires non-missing ratings on all six dimensions for at least one agro-dealer.[10] As a result, treatment effects on perceptions are estimated only for the final survey round and less precisely.

The model also predicts stronger updating among farmers with greater prior uncertainty (greater variance of priors). We examine this using prior agro-dealer seed-purchasing experience, proxied by whether the farmer used maize seed from an agro-dealer in the season preceding baseline. The subgroup estimates are 0.10 standard deviations among farmers without prior experience, compared to 0.09 in the full sample.

Because farmers without prior agro-dealer seed-purchasing experience are also likely to hold more biased priors, we additionally examine whether treatment effects are larger

[9] The presentation of results differs slightly from the pre-registered structure due to a reordering of outcomes, which modifies the construction of some pre-specified indices but leaves the main conclusions unchanged. Analyses adhering strictly to the pre-analysis plan remain publicly available in the project repository (see, e.g., the archived version linked here).

[10] A total of 1,534 farmers purchased seed from an agro-input shop for at least one plot at endline, so it is unsurprising to observe 1,664 farmers with non-missing ratings on all six dimensions for at least one agro-dealer.

Table 6: Treatment effects on farmers' perceived seed performance

	baseline	*full sample* *endline*		*non-purchasers* *endline*		*pessimists* *endline*	
	mean	treated	obs.	treated	obs.	treated	obs.
Index of farmer's seed ratings[1]	0.000	0.092^{+}	1664	0.102^{+}	996	0.116^{+}	934
	(0.637)	(0.054)		(0.055)		(0.062)	

Note: Column (1) reports the baseline mean and standard deviation below; column (2) reports the difference between treatment and control group and the standard error below; it is clustered at the level of randomization; column (3) reports the number of observations; columns (4) and (5) mirror this structure for the sub-sample of farmers who did not use high-yielding maize seed purchased from an agro-dealer on any plot at baseline; columns (6) and (7) mirror this structure for the sub-sample of farmers who reported to think that maize seed at agro-dealers is adulterated at baseline, **, *, and + denote significance at the 1%, 5%, and 10% levels; larger indices indicate more desirable outcomes. [1]The index of farmer's maize seed ratings of agro-dealers contains 6 ratings: general quality, yield, drought tolerance, pest/disease tolerance, time of maturity, germination. The ratings are aggregated at farmer level (one farmer rates multiple agro-dealers), then the index is computed.

among those with more pessimistic baseline beliefs (lower level of priors), proxied by whether the farmer believed that maize seed sold by agro-dealers is adulterated. The estimated treatment effect is 0.12 standard deviations among pessimistic farmers, compared with 0.09 in the full sample.

We interpret this pattern as suggestive evidence that beliefs update more strongly among farmers with more uncertain and pessimistic priors, consistent with the belief-updating channel. However, this subgroup analysis is exploratory, and the coefficients are imprecisely estimated, so we interpret these findings cautiously.

6.2 Adoption and sourcing decisions

Table 7 reports treatment effects on farmers' adoption and sourcing decisions. Consistent with the model's prediction that credible information about seed performance increases adoption, the Anderson (2008) index is significantly higher among treated farmers than among control farmers at both midline and endline, with effects significant at the 5% level.

Across all plots, treated farmers are 5.9 percentage points more likely to purchase high-yielding maize seed from an agro-dealer at midline, equivalent to 18% of the baseline mean. At endline, the estimated effect is three percentage points and no longer statistically significant, although treated farmers remain more likely than control farmers to adopt high-yielding maize varieties in both survey rounds: 3.5 percentage points at midline and 4.2 percentage points at endline. The response is concentrated on the extensive margin: treated farmers are more likely to source maize seed from agro-dealers, while quantities conditional on purchasing do not increase significantly.

We next examine sourcing decisions on a randomly selected maize plot. Treated farmers are 4.7 percentage points more likely at midline and 3.6 percentage points more likely at endline to obtain seed from an agro-dealer. These differences are accompanied by increased seed expenditure and reduced reliance on farmer-saved seed at midline. Effects on individual hybrid and OPV categories are positive but imprecisely estimated.[11]

We then assess whether the shift toward agro-dealer seed at the farmer level is reflected in agro-dealer business activity (Table 8). Outcomes include sales volume for the four most commonly purchased maize varieties (Longe 7H, Longe 10H, Longe 4, and Longe 5), average prices, implied revenue, the number of maize-seed customers served per day, and the number of maize varieties stocked.

The agro-dealer results are consistent with increased engagement at the farmer level. At midline, treated agro-dealers score significantly higher on the Anderson (2008) index, and their revenue is nearly 20% higher than that of control agro-dealers, significant at the 10% level. This revenue effect appears more consistent with higher sales volumes (28.4% relative to baseline) than with higher prices (2.3% relative to baseline), although neither component is individually statistically significant. At endline, the treatment effect on the business-activity index is significant at the 1% level, driven primarily by treated agro-dealers reporting 31% more customers than control agro-dealers, equivalent to roughly six additional customers per day.[12] Appendix A.7 reports business outcomes for two recently released maize varieties (Longe 10H and Longe 5), further supporting this pattern.

Taken together, the farmer and agro-dealer results show that the intervention shifts farmers toward purchasing seed from agro-dealers. These effects are consistent with the model's adoption prediction, but because all three channels can affect sourcing decisions, they do not identify a specific mechanism.

6.3 Productivity and harvest outcomes

We examine harvest-related outcomes on a randomly selected maize plot in Table 9. These include production, plot size, yield, the value of maize production at market prices net of seed costs, maize retained for seed in the following season, and maize sales outcomes.

The overall index shows a positive and statistically significant treatment effect at endline. Treated farmers achieve yields that are 50 kilograms per acre higher than those of control farmers, a 11% increase relative to the baseline mean and significant at the 1% level. The value of maize production net of seed costs is 66,000 UGX higher for

[11] We classify seed as hybrid or OPV if it corresponds to a recognized variety (e.g., Longe series, Bazooka, Panner, Wema, KH series) and was newly purchased rather than recycled or farmer-saved.

[12] Agro-dealers could display the certificate in their shops, and at endline it was visible in 56% of them. Ratings may therefore have been observed by non-sampled farmers, extending exposure beyond the study sample. Because randomization occurred at the catchment-area level, any such additional exposure remains within treated areas and does not threaten identification.

Table 7: Treatment effects on farmers' adoption and sourcing decisions

	baseline	*midline*		*endline*	
	mean	treated	obs.	treated	obs.
Farmer bought seed at agro-input shop for any plot†	0.325	0.059**	3145	0.031	3225
	(0.468)	(0.021)		(0.020)	
Amount of this seed farmer bought at agro-input shop in kg	9.519	-0.105	599	0.378	621
	(6.920)	(0.358)		(0.431)	
Farmer used high-yielding seed on any plot†	0.492	0.035^{+}	3206	0.042*	3282
	(0.500)	(0.020)		(0.020)	
Farmer bought seed at agro-input shop for specific plot†	0.330	0.047*	3153	0.036^{+}	3240
	(0.470)	(0.022)		(0.019)	
Farmer used farmer-saved seed on specific plot	0.579	-0.042^{+}	3153	-0.016	3240
	(0.494)	(0.022)		(0.020)	
Cost of seed used on specific plot in UGX§†	14078.272	0.499*	2848	0.350^{+}	2942
	(24654.685)	(0.235)		(0.209)	
Farmer used hybrid seed/OPV on specific plot^{1†}	0.432	0.035	2954	0.030	3047
	(0.495)	(0.023)		(0.023)	
Overall Anderson index	-0.013	0.087*	2854	0.086*	2978
	(0.899)	(0.042)		(0.039)	
Max. number of obs.			3407		3441

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

[1]For this variable, only non-recycled (newly purchased, not farmer-saved) seed counted as hybrid variety/OPV.

[§]Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

Table 8: Treatment effects on agro-dealers' business activity

	baseline	*midline*		*endline*	
	mean	treated	obs.	treated	obs.
Maize seed sales volume in kg^{1§†}	695.503	0.284	292	0.239	286
	(1497.183)	(0.227)		(0.253)	
Sales price of maize seed in UGX/kg^{2†}	4273.897	99.272	275	145.861	264
	(955.073)	(113.292)		(138.816)	
Revenue from maize seed in mln UGX^{3§†}	2.890	0.185^{+}	292	0.143	286
	(6.286)	(0.108)		(0.118)	
Number of maize seed customers per day§†	19.764	0.127	294	0.310**	288
	(20.689)	(0.101)		(0.112)	
Number of maize varieties in stock†	2.834	0.245	295	0.221	292
	(1.589)	(0.245)		(0.220)	
Overall Anderson index	0.031	0.197*	274	0.238**	270
	(0.610)	(0.092)		(0.082)	
Max. number of obs.			306		297

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

§Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

[1]Total quantity sold in the previous season of the four most widely used high-yielding maize varieties (Longe 7H, Longe 10H, Longe 4, and Longe 5), constructed from variety-level sales.

[2]Average start-of-season sales price across varieties, constructed from variety-level prices.

[3]Multiplying quantities sold by prices and summing across varieties.

treated farmers, approximately 18 USD at the time of the survey. Treated farmers also retain less maize for seed at midline, consistent with reduced reliance on farmer-saved seed. We find no effects on the quantity of maize sold or on revenue, suggesting that the yield gains do not translate into larger marketed surpluses and that additional output remains within the household.

A back-of-the-envelope calculation suggests that increased adoption alone is unlikely to fully explain the observed yield effect. Even if improved varieties yield 72% more than farmer-saved seed (Jovanovic and Ricker-Gilbert, 2025), the approximately five percentage point increase in agro-dealer sourcing would account for only a modest share of the observed 50 kg/acre yield gain.

The remaining effect likely reflects changes in how farmers use purchased seed. By reducing uncertainty about seed performance, the intervention may encourage greater investment in complementary inputs whose returns depend on seed quality, as documented by Emerick et al. (2016). Bohr et al. (2024) also show that correct beliefs about maize varieties can align fertilizer allocation. Consistent with this, treated farmers are about 5 percentage points more likely to use organic fertilizer at both midline and endline.

A related explanation is the reallocation of labor. Bulte et al. (2025) show that greater certainty about seed quality raises yields and reduces crop failure by inducing farmers to invest more effort. We find suggestive evidence consistent with this at midline: treated farmers are more likely to plant the recommended number of seeds per hill, indicative of increased production effort.

These harvest effects provide evidence that the intervention improves realized agricultural outcomes beyond beliefs and sourcing decisions. They are consistent with the model's prediction that credible seed-performance information increases productivity by shifting farmers toward higher-performing seed or by improving the performance of the seed they actually purchase. However, the observed yield gains are consistent with all three channels – belief updating, agro-dealer sorting, and reputational discipline – and therefore do not by themselves identify a specific mechanism.

6.4 Agro-dealer switching

We investigate whether the intervention induces farmers to reallocate purchases toward sellers of higher-performing seed. The model predicts that the agro-dealer sorting channel should generate switching, particularly where ratings reveal meaningful performance differences across agro-dealers. Unlike belief updating, which may induce farmers to shift away from farmer-saved seed toward agro-dealer seed markets, agro-dealer sorting requires farmers who already purchase from agro-dealers to switch across sellers.

We test this prediction by examining whether treated farmers are more likely to switch agro-dealers between adjacent seasons. The analysis is restricted to farmers who purchase maize seed from an agro-input shop in both seasons, since switching is only defined for them. Table 10 shows that switching is common: 46% of repeat buyers purchase from a different agro-dealer at midline than at baseline. The intervention

Table 9: Treatment effects on farmers' harvest on specific maize plot

	baseline	*midline*		*endline*	
	mean	treated	obs.	treated	obs.
Production in kg[†]	461.316	-21.153	2857	50.470**	2846
	(389.916)	(14.421)		(17.297)	
Area in acres	1.094	-0.003	3004	0.006	3066
	(0.655)	(0.029)		(0.038)	
Yield in kg/acre[†]	442.272	-24.662	2837	49.552**	2836
	(303.884)	(17.203)		(18.008)	
Production minus cost of seed in UGX[1]	301696.183	-3898.858	2409	62288.963**	2494
	(281903.847)	(9320.232)		(22847.887)	
Amount kept as seed in kg[§]	14.506	-0.188*	2931	0.036	2861
	(18.530)	(0.092)		(0.104)	
Amount sold in kg[§†]	195.295	-0.201	3063	0.173	3137
	(297.545)	(0.124)		(0.173)	
Sales price in UGX/kg	506.954	33.027*	610	12.614	639
	(139.389)	(14.244)		(41.238)	
Revenue in UGX[§†]	97.783	-0.393	3058	0.355	3109
	(156.538)	(0.257)		(0.363)	
Overall Anderson index	-0.014	-0.061	2914	0.104*	2871
	(0.787)	(0.040)		(0.043)	
Max. number of obs.			3407		3441

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; [†] indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

[§]Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

[1]Production in UGX, measured as the number of maize bags harvested, multiplied by the market value per bag in UGX, minus the cost of maize seed used in UGX, calculated as the quantity of seed used in kg, multiplied by the cost per kg of seed in UGX.

increases switching by 6.2 percentage points at midline, significant at the 10% level, but the estimate is smaller and no longer statistically significant at endline. It may induce some short-run reallocation across sellers, but the effect does not persist.

An exploratory test of the agro-dealer sorting channel examines whether treated farmers reallocate purchases toward better-rated agro-dealers. Restricting the sample to farmer-agro-dealer pairs in which the farmer had previously purchased from the agro-dealer, we estimate the following linear probability model for whether farmer f repurchases from agro-dealer d :

$$\text{repurchase}_{fd} = \alpha + \beta_1 T_j + \beta_2 s_d + \beta_3 (T_j \times s_d) + \varepsilon_{fd}$$

where T_j is a binary indicator for treatment assignment, s_d is the agro-dealer's aggregated peer rating – the information disseminated by the intervention, and standard errors are clustered at the level of catchment area j.

The sorting channel predicts a positive interaction: treated farmers should be more likely to repurchase from agro-dealers with higher ratings. We find no evidence of such a pattern. The interaction coefficient is small, negative, and statistically insignificant ($\beta_3 =$ -0.038, standard error 0.028; 2,171 farmer-agro-dealer pairs from 1,645 farmers).

A complementary farmer-level analysis reaches the same conclusion. Treated farmers purchase from agro-dealers with ratings only 0.05 points higher than those chosen by control farmers (3.65 vs. 3.60), a difference that is not statistically significant. Thus, treated farmers are not more likely to repurchase from higher-rated agro-dealers.

Taken together, these findings provide no support for agro-dealer sorting as the primary mechanism. The distribution of agro-dealer ratings helps explain why. Ratings show little variation: at endline, the mean is 3.6 with a standard deviation of 0.3, the interquartile range spans 3.44 to 3.84, and 98% of agro-dealers are rated above the midpoint of the scale. This matches the condition under which the model says the sorting channel should be weak: when ratings are similar across agro-dealers, they contain little agro-dealer-specific information (Section 2.3), and our assumption that intrinsic seed performance is common across agro-dealers, so that any cross-dealer variation arises from differences in handling and storage effort (Section 2.1).[13]

This compression limits the power of any sorting test, but it is also substantively informative: when the farmers who rate regard essentially all agro-dealers as acceptable, the binding constraint is not which agro-dealer to choose, but whether purchasing seed from an agro-dealer is worthwhile at all – consistent with the treatment operating through farmers' beliefs about agro-dealer seed performance in general, rather than through sorting across agro-dealers. This is in line with Hasanain, Khan, and Rezaee (2023), who find no evidence that farmers switch to higher-performing providers in response to an information platform for artificial livestock insemination services in Pakistan.

[13] For evidence on the informativeness of the rating data, see Subsection 5.4 and Appendix A.6.

Table 10: Treatment effects on farmers' switching behavior

	midline	*midline*		*endline*	
	mean	treated	obs.	treated	obs.
Farmer switched to different shop	0.457	0.062^{+}	1270	0.031	1410
	(0.498)	(0.033)		(0.033)	
Max. number of obs.[1]			1270		1410

Note: Column (1) reports midline means and standard deviations below (because this variable was not collected at baseline); columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels.
[1]Comparisons are restricted to farmers who purchased maize seed from an agro-input shop for any plot at mid- or endline and during the season before that (i.e., base- or midline).

6.5 Agro-dealer responses and seed performance

We investigate whether the intervention induces agro-dealers to change their behavior. In the model, public ratings increase agro-dealers' reputational return to selling well-performing seed under the reputational-discipline channel. This has two empirical implications. First, agro-dealers may respond through improvements in seed handling, customer service, or quality signaling. Second, if these responses affect the seed itself, they should translate into measurable improvements in objective seed performance.

Table 11 reports treatment effects on agro-dealer responses. We construct two indices from enumerator observations of seed handling and storage: labor-intensive practices, such as not storing seed in open containers, and capital-intensive practices, such as having a leak-proof roof. Neither index is affected by the intervention, providing no evidence that agro-dealers change the practices most directly related to seed preservation.

The intervention does, however, generate suggestive evidence of customer-facing responses. We measure service provision using both agro-dealer and farmer reports. The agro-dealer-reported service index – including, e.g., recommendations about complementary inputs – is higher among treated agro-dealers at endline, while a parallel farmer-reported service index is higher at midline. Although the timing differs, both measures point to improved service provision. We also find a positive effect on quality signaling – activities that convey credibility or professionalism to customers, such as membership in a professional association – with the index higher among treated agro-dealers at endline. Together, these patterns suggest that agro-dealers respond to the intervention along service and signaling margins.

We then examine whether these agro-dealer responses translate into measurable improvements in seed performance. Table 12 reports treatment effects on several seed-performance indicators. Enumerators purchased an original, sealed bag of maize seed from agro-dealers and assessed moisture content, packaging integrity, and packaging information. Consistent with the absence of changes in seed handling and storage

Table 11: Treatment effects on agro-dealers' practices

	baseline	*midline*		*endline*	
	mean	treated	obs.	treated	obs.
Index of labor-intensive seed handling practices[1]†	0.010	0.099	285	0.074	274
	(0.484)	(0.065)		(0.068)	
Index of capital-intensive seed handling practices[2]†	0.000	0.000	270	0.057	265
	(0.559)	(0.072)		(0.083)	
Index of services (self-reported by agro-dealers)[3]†	0.000	0.066	243	0.110*	247
	(0.382)	(0.060)		(0.052)	
Index of services (reported by farmers)[4]†	-0.027	0.301**	259	0.086	271
	(0.583)	(0.069)		(0.084)	
Index of quality signaling practices[5]†	0.002	0.153	267	0.265+	255
	(0.686)	(0.103)		(0.147)	
Overall Anderson index	0.034	0.375**	162	0.178+	177
	(0.537)	(0.113)		(0.106)	
Max. number of obs. for dealer survey outcomes			306		297

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

[1]The index of labor-intensive seed handling practices contains 6 variables: whether seed is stored in dedicated area, in correct lighting, on correct surface, not in open containers, whether shop has no pest problem, cleanness and professionality rating by enumerator.

[2]The index of capital-intensive seed handling practices contains 5 variables: whether roof is leak-proof, roof is insulated, walls are insulated, shop is ventilated, expired seed is handled correctly. All variables were observed or verified by enumerators, except the last, which is self-reported.

[3]The index of services (self-reported) contains 8 variables: whether shop offers explanations, complementary input recommendations, extension/training, discounts for larger quantities, credit, did not receive seed related customer complaint, accepts mobile money, stocks small (not repackaged) seed package.

[4]The index of services (reported by farmers) contains 7 variables: whether shop offers refund/insurance, credit, training/advice, delivery, after-sales service, accepts different payment methods, sells small quantities. Farmers' responses are aggregated at dealer level, then the index is computed.

[5]The index of quality signaling practices includes 5 variables: whether shop displays official certificate, is registered with UNADA, is a member of another professional association, holds a trading license issued by local government, number of shop inspections.

practices most directly related to seed preservation, we find no significant treatment effects on any of these indicators.

Because not all sampled agro-dealers had maize seed in stock at the time of the survey, the seed-performance sample is smaller than the main sample. To assess whether the null result reflects limited statistical precision, Appendix A.8 reports a power analysis for moisture content, a pre-registered objective indicator of seed performance. The analysis shows that the estimated endline effect is small relative to the variation in the data and would remain difficult to detect even with a substantially larger sample. As a complementary check, Table A7 examines additional seed-performance proxies available for larger samples and again finds no treatment effects. We therefore find no evidence of detectable improvements in the measured seed-performance indicators.

Taken together, the supply-side results provide some evidence that agro-dealers respond to the intervention, primarily along service and signaling margins. However, we find no detectable improvements in seed handling, storage, or measured seed performance. The evidence therefore points to a limited seller response, but provides no support for reputational discipline as the primary channel, which would require reputational pressure to translate into improvements in the objective performance of seed sold by agro-dealers.

7 Conclusion

This paper studies how aggregating and disseminating peer experiences affects adoption and productivity when farmers face uncertainty about the performance of agricultural inputs. We designed an intervention that collected farmers' experiences with maize seed purchased from agro-dealers, aggregated these into agro-dealer-level ratings, and disseminated them to farmers and agro-dealers in a randomized experiment.

A conceptual model organizes three channels through which the intervention may affect outcomes: belief updating about agro-dealer seed, sorting across agro-dealers, and reputational discipline. Measures of farmers' perceptions, adoption, switching, and harvests, alongside agro-dealer responses and seed-performance indicators, allow us to assess which mechanism is most consistent with the observed pattern of results.

The intervention affects farmers' perceptions, behavior, and harvests. Treated farmers hold more favorable perceptions of agro-dealer seed performance. The intervention also increases adoption and purchases from agro-dealers. Treated farmers achieve higher yields and higher net production values. These gains are consistent with credible seed-performance information shifting farmers toward better-performing seed, changing how purchased seed is used, or improving the performance of seed at agro-dealers. However, because adoption and productivity can increase under all three channels, these outcomes alone do not identify the underlying mechanism.

The overall pattern of evidence points most strongly to belief updating. Switching across agro-dealers increases modestly at midline but does not persist, providing limited support for agro-dealer sorting as the primary channel. Agro-dealers respond along

Table 12: Treatment effects on agro-dealers' seed performance indicators

	baseline	*midline*		*endline*	
	mean	treated	obs.	treated	obs.
Moisture in %†	13.564	-0.122	175	-0.220	261
	(1.482)	(0.144)		(0.197)	
Bag shows packaging date†	0.689	0.050	179	0.035	265
	(0.464)	(0.072)		(0.064)	
Shelf-life in days[1]†	60.951	-8.272	164	6.352	240
	(40.960)	(20.869)		(8.289)	
Seed is in original undamaged bag†	0.940	0.002	179	0.051	265
	(0.238)	(0.046)		(0.055)	
Bag shows lot number†	0.508	-0.001	179	0.027	265
	(0.501)	(0.107)		(0.064)	
Overall Anderson index	0.065	0.108	160	0.108	236
	(0.364)	(0.103)		(0.090)	
Max. number of obs.[2]			179		265

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

[1]Days since the packaging date or, if the bag does not show the packaging date, days since the expiry date minus 6 months.

[2]The comparisons were only made for agro-dealers where the enumerator was able to buy a bag of maize seed at mid- or endline. Also, we do not control for the baseline values of the outcome variables in the entire table because only 144 of the 179 dealers who had seed at midline also had seed at baseline and only 183 of the 265 dealers who had seed at endline also had seed at baseline, so that controlling for baseline values would reduce the sample sizes drastically.

service and signaling margins, yet we detect no improvements in seed handling, storage, or measured seed-performance, providing limited support for reputational discipline as the primary channel.

The findings suggest that the binding constraint was not farmers' inability to identify the best seller, but uncertainty about whether agro-dealer seed was worth purchasing at all. This distinction has important implications. In markets where buyers already participate but face difficulty choosing among sellers, ratings may operate primarily through sorting across suppliers. In markets where many potential buyers remain outside the formal market because input performance is uncertain, broad dissemination of credible information may instead increase market participation itself.

The policy implication is that experience-based information systems can complement formal quality assurance where regulation, certification, or private branding provide incomplete signals to farmers. By aggregating user experiences, such decentralized systems may reduce informational barriers to adoption and increase engagement with input markets. At the same time, the results caution against assuming that transparency alone generates substantial supply-side quality improvements. Strengthening seller incentives and improving input performance may require more direct intervention.

As smartphone access expands among smallholder farmers, digital systems could reach them at lower cost and larger scale. An important question for future research is whether digital delivery can preserve the credibility, local relevance, and informational value of an offline intervention while also generating stronger incentives for agro-dealers to improve input quality.

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A Appendix

A.1 Context on seeds and their quality

In this study, the term "high-yielding maize varieties" refers to both open-pollinated varieties (OPVs) and hybrids that have been genetically improved through public or private sector breeding programs. While the difference between these two types of maize is based on their reproductive biology, there are also economic implications (Morris, Rusike, and Smale, 1998; Spielman and Kennedy, 2016). We explain the terminology and our rationale for using this terminology below, recognizing that our explanation is a simplification of much more complex biological phenomena.

Maize hybrids tend to confer a significant advantage in terms of vigor – robust growth, high yields, and uniform yield performance, even under suboptimal conditions. This is due to the expression of heterosis that results from crossing distinct inbred parent lines to produce genetically superior progeny (F1) that is planted by the farmer. However, this vigor declines dramatically if farmers save and plant grain that was set aside from harvest. Instead, the farmer must purchase fresh (F1) seed each season to continue to realize the benefits of heterosis. The combination of vigor and purchase frequency can incentivize public investment in hybrid maize improvement, seed production, and marketing.

OPVs, on the other hand, are typically less vigorous than hybrids. Although they are also less susceptible to genetic degeneration and can be saved and planted across several generations without loss of vigor, their yield performance is typically lower than hybrids. Moreover, OPV seed saving practices – and thus less frequent purchases of fresh seed – tend to reduce these private investment incentives.

Relatedly, we use the term "farmer-saved seed" in this study to refer to seed that is of an unknown progeny or quality. They may be seed saved from the harvest of F1 hybrids or OPVs, and they may be n generations old. They may have been stored under suboptimal conditions between seasons, exposed to moisture, pests, or diseases, and, as a result of both genetic and physical factors, perform poorly relative to fresh hybrid or OPV seed purchased from an agro-input shop.

As highlighted in a recent review by Michelson et al. (2023), seed quality is complex and multi-dimensional due to the living nature of seeds, encompassing three essential dimensions. First, analytical purity measures the proportion of a seed sample that belongs to the intended species. Second, varietal purity ensures the seed is of the expected variety, possessing the claimed genetic characteristics. Third, viability and performance assess the seed's capacity to germinate and grow with sufficient vigor.

This study focuses on the viability and performance dimension of maize seed quality. From a farmer's perspective, this dimension is particularly salient, as it represents a crucial outcome that farmers observe during germination, early crop growth, and harvest, although it can be difficult to attribute realized outcomes to seed performance alone. Moreover, while some storage and handling practices, such as seed mixing or mislabeling, may affect varietal purity, the practices examined in this study primarily influence seed viability and performance.

Objectively assessing maize seed performance is complex, but measuring the moisture content of a seed sample is the most common and practical test. Moisture is a powerful predictor, as it directly affects germination outcomes. Afzal et al. (2017), e.g., report germination rates of approximately 85% for seeds with a moisture content of 12.7%, compared to just 50% for the same seeds at 15.6%. Prolonged exposure to high humidity rapidly degrades seed viability by increasing susceptibility to damage from pathogens, insects, and physical handling (Michelson et al., 2023). The moisture content of a seed sample reflects recent storage conditions – whether the seeds were kept in airtight packages or exposed to open air. Consulted experts emphasized a widely used rule of thumb: each one-percentage-point increase in moisture content halves the seed's shelf life.

While moisture measurement is standard practice among seed companies and crop inspectors and provides a valuable indication of seed performance, it is inherently one-dimensional. Additional quality assessments include germination tests conducted in controlled chambers; vigor tests under the growing conditions that may occur in the field; grow-out tests, which evaluate genetic purity through morphological analysis; seed health testing which measures the presence of pathogens and diseases, and genotyping (DNA fingerprinting), which verifies genetic purity using single nucleotide polymorphism analysis.

A.2 Rating computation and dissemination

What if a treated agro-dealer does not receive a single rating? If a treated agro-dealer is not rated by any farmer – e.g., because no sampled farmer had personally purchased seed there or knew someone who had – we informed farmers that we had no information about the shop while still mentioning its existence. In the first round, 16 out of 193 treated agro-dealers were not rated by any farmer.

Should more ratings lead to better ratings? Some agro-dealers were not rated by any farmer in the first round, while others received up to 22 ratings. If agro-dealer A is rated by ten farmers with an average of 3.5 and agro-dealer B is rated by one farmer with a rating of 3.6, agro-dealer B is treated as having the higher rating.

Should ratings reflect catchment area performance? Ratings were highly concentrated at the upper end of the scale, with fewer than 9% of agro-dealers receiving a rating below three out of five. Ratings are therefore transformed into quintiles based on their distribution. To avoid conveying an overly negative signal, the quintile labels were chosen to carry positive connotations.

Are female agro-dealers rated worse than male agro-dealers? Significant differences were observed between the ratings of female agro-dealers (41% of the sample) and male agro-dealers (59%), even after controlling for potential confounders such as education and multiple performance indicators. These differences are unlikely to reflect true disparities in performance and instead appear to stem from biased perceptions, likely driven by discrimination (De, Miehe, and Van Campenhout, 2024). To mitigate potential harm from the intervention, the ratings of female agro-dealers were adjusted accordingly. Specifically, each seed performance attribute was regressed on a gender dummy, and the resulting coefficients were added to the initial ratings of female agro-dealers.

A.3 Agro-dealer rating report

Shifrahz Agro Farm supply
scores

4.4 out of 5

This was based on 8 reviews.

This score means that farmers in the area think that the quality of the maize seed this shop sells is:

Excellent!

SeedAdvisor certificate 2022

Hi there!

As a business, you know that your customers' opinion is very important! Your sales and profits depend on what customers think about the quality of the products you sell.

We visited farmers in your area and asked them about their experience with maize seed that they bought in shops such as yours. We have used this information to compute a score for you. In this way, you get an idea of what customers think about the maize seed you sell, and how it compares to other agro-input shops in your area. This gives you the opportunity to improve your business even more!

The overall score indicated on the certificate combines different components. You can find separate ratings in the table below. For instance, we asked what customers think of maize seed quality in general. But we also asked customers about their experiences with respect to yield, drought and disease resistance, time to mature, and germination, keeping in mind what was advertised when they bought the seed.

Check your scores carefully and see where you can do better! You can also compare your own results to what customers think of the quality of maize seed sold by other agro-input dealers in your area.

	you	other input dealers
What do customers say about overall seed quality?	★★★☆☆	★★★⯪☆
Do customers think yield of maize seed is as advertised?	★★★★★	★★★⯪☆
Do customers think maize seed is as drought resistant as advertised?	★★★★☆	★★★⯪☆
Do customers think maize seed is as disease resistant as advertised?	★★★★⯪	★★★⯪☆
Do customers think maize seed matures as advertised?	★★★★⯪	★★★☆☆
What do customers think about maize seed germination?	★★★★⯪	★★★☆☆

Tip: Display this certificate in your shop to inform customers about your score. You will be scored again in July 2022, so keep standards up!

A.4 Alternative index construction

We recompute all summary indices following Kling, Liebman, and Katz (2007) as a robustness check to the Anderson (2008) indices used in the main analysis. For farmers' perceptions of seed performance, the treatment effect becomes statistically insignificant when the index is constructed following Kling, Liebman, and Katz (2007). Results for farmers' adoption and sourcing decisions remain robust. Results for agro-dealers' business activity (general, Longe 10H, and Longe 5) are largely robust.

For farmers' harvest outcomes, the positive endline effect remains robust. At midline, however, the Kling, Liebman, and Katz (2007) index yields a significant negative coefficient. The harvest outcomes included in the index are mechanically related (e.g., yield, production, and revenue). The Anderson (2008) method down-weights highly correlated components, whereas Kling, Liebman, and Katz (2007) assigns equal weights and treats these outcomes as independent, which can amplify the influence of redundant information and generate a negative treatment effect. That is why we rely on the Anderson (2008) method, as pre-registered in our pre-analysis plan, and use Kling, Liebman, and Katz (2007) only as a robustness check.

Results for agro-dealers' practices remain robust, as do all null results regarding agro-dealers' seed performance.

Table A1: Anderson (2008) vs. Kling, Liebman, and Katz (2007) indices

	Anderson		Kling et al.	
	midline	*endline*	*midline*	*endline*
Farmers' perceived seed performance		0.092^{+}		0.088
		(0.054)		(0.056)
Farmers' adoption and sourcing decisions	0.087^{*}	0.086^{*}	0.096^{*}	0.076^{*}
	(0.042)	(0.039)	(0.041)	(0.038)
Agro-dealers' business activity (general)	0.197^{*}	0.238^{**}	0.189	0.247^{+}
	(0.092)	(0.082)	(0.125)	(0.132)
Agro-dealers' business activity (Longe 10H)	0.029	0.217^{**}	0.060	0.232^{**}
	(0.070)	(0.057)	(0.082)	(0.076)
Agro-dealers' business activity (Longe 5)	0.012	0.152^{*}	0.105	0.160^{+}
	(0.062)	(0.058)	(0.089)	(0.081)
Farmers' harvest	-0.061	0.104^{*}	-0.093^{*}	0.090^{+}
	(0.040)	(0.043)	(0.046)	(0.047)
Agro-dealers' practices	0.375^{**}	0.178^{+}	0.272^{*}	0.238^{*}
	(0.113)	(0.106)	(0.118)	(0.118)
Agro-dealers' seed performance indicators	0.108	0.108	0.065	0.064
	(0.103)	(0.090)	(0.111)	(0.087)
Agro-dealers' seed performance (alternative proxies)	0.066	0.060	0.062	0.059
	(0.100)	(0.087)	(0.103)	(0.087)

Note: Columns (1) and (2) report differences between treatment and control groups, with standard errors below (clustered at the level of randomization), if the overall indices are constructed following Anderson (2008); columns (3) and (4) report differences between treatment and control groups, with standard errors below (clustered at the level of randomization), if the overall indices are constructed following Kling, Liebman, and Katz (2007); **, *, and + denote significance at the 1%, 5%, and 10% levels; larger indices indicate more desirable outcomes.

A.5 Differential attrition tests

To examine differential attrition, we regress the pre-registered balance variables (see Table 4) on a binary variable indicating whether an agro-dealer left the sample, a binary variable indicating whether an agro-dealer belonged to the control group, and their interaction. Column (2) reveals that agro-dealers who left the sample at midline operated closer to the nearest tarmac road – the only significant difference between those who left and those who stayed when attrition was most pronounced (see Table 5). By endline, however, those who left were younger, more likely to be female, had less established businesses, were more likely to operate specialized shops, and still tended to be closer to a tarmac road, see column (5). This suggests that agro-dealers who left differed from those who remained, regardless of exposure to the treatment, possibly reflecting greater vulnerability during the COVID-19 period, leading to business closures. Columns (3) and (6) show minimal differences between control and treated agro-dealers, consistent with Subsection 5.2. Examining the interaction between the indicators reveals few differences between agro-dealers who left the sample *and* belonged to the control group and those who did not, suggesting that differential attrition is unlikely to bias our estimates.

Table A2: Differential attrition tests - Agro-dealer

	baseline		*midline*			*endline*	
	mean	attrited	belongs to control	attrited and control	attrited	belongs to control	attrited and control
Respondent's age in years	32.43	-2.50	2.68^{+}	-1.29	-5.59**	1.59	5.45^{+}
	(11.49)	(2.25)	(1.40)	(3.04)	(1.27)	(1.40)	(3.04)
Resp. is male	0.59	-0.17	0.05	-0.12	-0.34**	-0.02	0.28^{+}
	(0.14)	(0.06)	(0.17)	(0.49)	(0.10)	(0.06)	(0.14)
Resp. finished primary education	0.92	0.01	0.01	-0.02	0.04	0.02	-0.04
	(0.27)	(0.08)	(0.04)	(0.10)	(0.05)	(0.04)	(0.08)
Resp. owns shop	0.55	-0.15	-0.01	0.01	-0.05	-0.05	0.14
	(0.50)	(0.17)	(0.06)	(0.20)	(0.16)	(0.07)	(0.20)
Resp. received training on maize seed handling	0.53	0.14	-0.11	-0.18	0.10	-0.09	-0.19
	(0.50)	(0.11)	(0.07)	(0.16)	(0.10)	(0.07)	(0.15)
Resp. knows how to store seed after repackaging	0.27	0.13	-0.08	-0.10	0.03	-0.09	0.03
	(0.44)	(0.17)	(0.06)	(0.19)	(0.12)	(0.06)	(0.15)
Dealer's distance to nearest tarmac road in km	6.56	-5.37**	1.68	2.63	-3.38^{+}	1.67	0.92
	(10.39)	(1.37)	(2.33)	(2.63)	(1.74)	(2.35)	(2.93)
Dealer only sells farm inputs	0.74	0.11	0.00	-0.18	0.23**	0.01	-0.27*
	(0.44)	(0.10)	(0.07)	(0.14)	(0.07)	(0.07)	(0.12)
Years since dealer establishment	5.34	-1.93	0.07	-0.35	-2.99*	-0.08	0.60
	(6.30)	(1.26)	(0.81)	(1.78)	(0.96)	(0.82)	(1.51)
Number of customers per day	41.49	-7.99	-5.02	-5.48	-8.05	-5.35	-4.79
	(46.49)	(8.66)	(7.50)	(10.59)	(7.01)	(7.73)	(9.53)
Quantity of maize seed sold in kg	695.50	234.18	-144.71	-847.11	125.10	-88.44	-551.67
	(1497.18)	(833.31)	(376.10)	(875.12)	(459.81)	(228.49)	(483.08)
Maize seed lost/wasted last season in kg	3.50	2.73	-2.22	-3.06	3.91	-1.68	-5.89
	(18.65)	(8.06)	(2.35)	(8.16)	(9.08)	(2.59)	(9.21)
Dealer has problem with pests	0.65	0.16	0.08	-0.36*	-0.02	0.04	-0.08
	(0.48)	(0.11)	(0.06)	(0.17)	(0.09)	(0.06)	(0.14)
Dealer stores maize seed in open containers	0.16	-0.05	-0.09^{+}	0.09	-0.05	-0.09^{+}	0.09
	(0.36)	(0.10)	(0.05)	(0.12)	(0.10)	(0.05)	(0.12)
Dealer received seed related customer complaint	0.64	-0.03	-0.04	-0.14	-0.11	-0.07	0.05
	(0.48)	(0.12)	(0.05)	(0.17)	(0.12)	(0.06)	(0.16)
Moisture in bag of maize seed in %	13.56	-0.42	0.10	0.27	-0.45	0.10	0.48
	(1.44)	(0.52)	(0.27)	(0.66)	(0.31)	(0.26)	(0.56)

Note: Column (1) reports sample means at baseline and standard deviations below; columns (2) and (5) report differences between agro-dealers who attrited and those who did not, and standard errors below; columns (3) and (6) report differences between control and treatment agro-dealers, and standard errors below; columns (4) and (7) report differences between agro-dealers who attrited *and* belong to the control group and those who do not, and standard errors below; all standard errors are clustered at the level of randomization; **, *, and + denote significance at the 1%, 5%, and 10% levels.

A.6 Agro-dealer rating patterns

Comparison of rating dimensions We asked farmers to evaluate maize seed sold by an agro-dealer across six dimensions: General quality, Yield as advertised, Drought tolerance as advertised, Pest/disease tolerance as advertised, Speed of maturing as advertised, and Germination. Figure A1 presents the distribution of farmers' ratings for these attributes along with their mean values.

The figure reveals variation across attributes, with some showing more concentrated distributions and others exhibiting greater dispersion. The attributes most strongly correlated with the *General quality* rating are *Yield as advertised* (0.54) and *Germination* (0.38). These two attributes also have the smallest mean absolute differences (0.47 for *Yield as advertised* and 0.55 for *Germination*), suggesting that these are the attributes farmers primarily consider when assessing overall performance.

Comparison of ratings by purchase channel Farmers could rate an agro-dealer if they met one of two conditions: either they had purchased seed from the agro-dealer themselves or they knew someone who had. Ideally, only farmers with firsthand experience would provide ratings, but restricting responses in this way risked generating too few observations for a reliable signal. While this decision was made at the experiment's design stage and cannot be revised ex post, we can compare endline ratings between farmers who purchased seed from the agro-dealer and those who did not but knew someone who had.

Fortunately, the vast majority (93%) of ratings came from farmers who had purchased seed, while only 7% were provided by those without direct purchase experience. This alleviates concerns that ratings are primarily driven by reputation or hearsay effects. While the mean ratings differ slightly (3.6 for farmers who bought seed vs. 3.7 for those who did not), the density plots are similar (see Figure A2).

Correlating ratings and performance indicators There is concern that smallholder farmers may base their ratings of agro-input shops on factors such as customer service rather than seed performance or rely on the agro-dealer's reputation or personal relationship with them. To address this, we test whether the ratings – specifically, the index of endline ratings computed following Anderson (2008) – correlate with objective indicators of seller and seed performance, also measured at endline.

We find that specialized agro-input shops, which exclusively sell farm inputs, receive higher ratings, as do agro-dealers with better seed handling practices, objectively observed by our enumerators (Table A3). Also, farmers not only rated seeds purchased from agro-input shops but also those used on randomly selected maize fields, using the same questions. These seed ratings are positively correlated with crop yields (Table A4).

While alternative explanations for these significant correlations exist, the overall pattern supports our claim that farmers are assessing seed performance. The ratings therefore appear to provide an informative signal, although farmers' perceptions may not be perfectly accurate given the many factors influencing agricultural production.

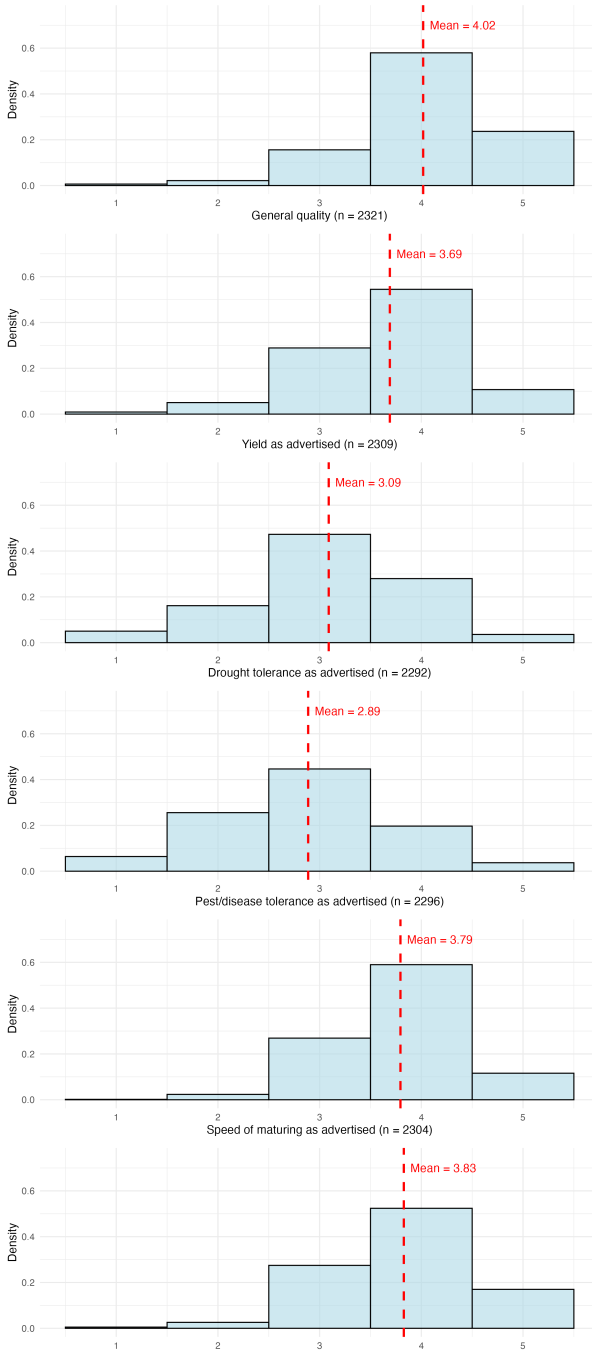

Figure A1: Comparison of rating dimensions at endline

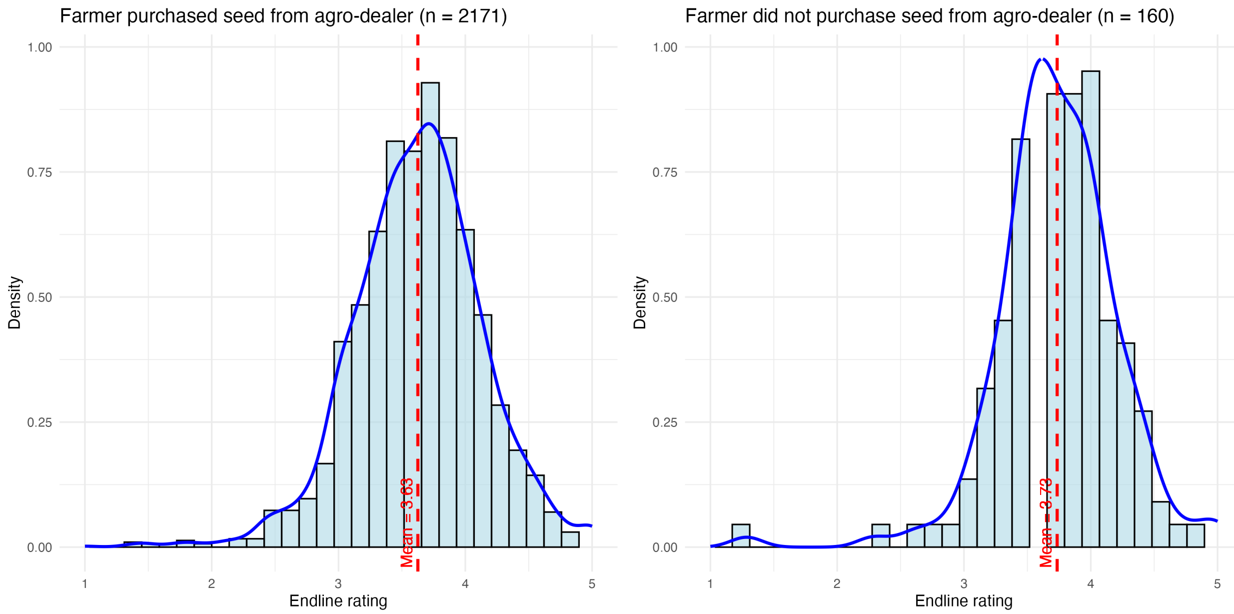

Figure A2: Comparison of ratings by purchase channel

Table A3: Correlating ratings and performance indicators - Agro-dealer level

	Ratings
Agro-dealer only sells farm inputs	0.205^{**}
	(0.078)
Index of labor-intensive seed handling practices[1]	0.164^{+}
	(0.091)
Index of capital-intensive seed handling practices[2]	0.156^{+}
	(0.081)
Index of all seed handling practices[3]	0.205^{+}
	(0.116)
Agro-dealer received seed related complaint from customer	-0.108
	(0.093)
Shop received warning after inspection	-0.034
	(0.086)

Note: [1]The index of labor-intensive seed handling practices contains 6 variables: whether seed is stored in dedicated area, in correct lighting, on correct surface, not in open containers, whether shop has no pest problem, cleanness and professionality rating by enumerator.
[2]The index of capital-intensive seed handling practices contains 6 variables: whether roof is leak-proof, roof is insulated, walls are insulated, shop is ventilated, shop displays official certificate, expired seed is handled correctly.
[3]The index of all seed handling practices contains 12 variables: the ones included in the index of capital-intensive practices and the ones included in the index of labor-intensive practices.

Table A4: Correlating ratings and yield on randomly selected field - Farmer level

	Yield in kg/acre$_{\text{endline}}$
Index of farmer's seed ratings$_{\text{midline}}$[1]	138.634^{**}
	(8.322)
Index of farmer's seed ratings$_{\text{endline}}$[1]	38.625^{**}
	(9.179)

Note: [1]The index of farmer's seed ratings contains 6 ratings: general quality, yield, drought tolerance, pest/disease tolerance, speed of maturing, and germination.

A.7 Treatment effects on agro-dealers' business activity

We examine treatment effects on agro-dealer business activity for two maize varieties: the most recently released hybrid, Longe 10H (Table A5) and the most recently released OPV, Longe 5 (Table A6). In addition to the general business outcomes described in Subsection 6.2, we include stock indicators. Agro-dealers reported how much seed of each variety was carried over from the previous season, how much was procured from suppliers, how much was lost or wasted, and how often stock-outs occurred.

Higher demand induced by the intervention should increase turnover, reducing carryover and increasing procurement of fresh seed. Higher turnover may also reduce losses and stock-outs.

At midline, we find no significant effects. By endline, however, all coefficients move in the expected direction, and the combined Anderson (2008) index shows a positive and statistically significant effect. This pattern is consistent with increased farmer engagement translating into higher agro-dealer business activity.

Table A5: Treatment effects on agro-dealers' business activity (Longe 10H)

	baseline	*midline*		*endline*	
	mean	treated	obs.	treated	obs.
Sales volume in kg§†	288.384	0.236	256	0.352	242
	(727.049)	(0.204)		(0.239)	
Sales price in UGX/kg§†	6220.899	-0.013	194	0.010	187
	(1153.371)	(0.026)		(0.029)	
Revenue in mln UGX§†	1.625	0.130	255	0.173	241
	(3.839)	(0.123)		(0.136)	
Amount carried over in kg^{1§†}	2.679	0.090	262	-0.034	250
	(12.137)	(0.215)		(0.134)	
Amount shop bought from provider in kg§†	294.672	0.206	257	-0.022	243
	(741.810)	(0.213)		(0.253)	
Amount lost/wasted in kg§†	0.036	0.019	257	-0.038	243
	(0.405)	(0.097)		(0.041)	
Number of times per month shop ran out§†	1.039	-0.045	192	-0.205	185
	(1.575)	(0.133)		(0.136)	
Overall Anderson index	0.080	0.029	244	0.217**	233
	(0.437)	(0.070)		(0.057)	
Max. number of obs.[2]			268		254

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

§Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

[1]Amount carried over is low at baseline, as many agro-dealers report carrying no seed over from the previous season.

[2]The comparisons were only made for agro-dealers that had Longe 10H in stock at mid- or endline.

Table A6: Treatment effects on agro-dealers' business activity (Longe 5)

	baseline	*midline*		*endline*	
	mean	treated	obs.	treated	obs.
Sales volume in kg§†	389.492	0.304	261	0.316	259
	(716.556)	(0.216)		(0.230)	
Sales price in UGX/kg§†	3114.463	-0.015	249	0.013	241
	(409.328)	(0.016)		(0.022)	
Revenue in mln UGX§†	1.193	0.111	261	0.114	258
	(2.175)	(0.096)		(0.105)	
Amount carried over in kg^{1§†}	4.312	-0.092	270	-0.004	263
	(19.088)	(0.306)		(0.155)	
Amount shop bought from provider in kg§†	431.451	0.253	262	0.289	260
	(803.696)	(0.215)		(0.235)	
Amount lost/wasted in kg§†	1.756	0.031	266	-0.055	261
	(10.173)	(0.128)		(0.058)	
Number of times per month shop ran out§†	0.839	0.086	248	-0.054	237
	(1.509)	(0.101)		(0.126)	
Overall Anderson index	0.039	0.012	256	0.152*	252
	(0.401)	(0.062)		(0.058)	
Max. number of obs.[2]			275		269

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.

§Due to the skewness of this variable, the regression was run after an inverse hyperbolic sine transformation. Coefficient estimates can therefore be interpreted as percentage changes. The baseline mean column shows the untransformed variable.

[1]Amount carried over is low at baseline, as many agro-dealers report carrying no seed over from the previous season.

[2]The comparisons were only made for agro-dealers that had Longe 5 in stock at mid- or endline.

A.8 Treatment effects on agro-dealers' seed performance

Moisture content – a pre-registered, objectively measured indicator of seed performance – shows a statistically insignificant negative treatment effect. Because not all agro-dealers had maize seed in stock, the realized sample size is smaller than planned: 232 observations at baseline, 179 at midline, and 265 at endline, compared with a target of 348.

To assess whether the null result reflects a true absence of effect or insufficient statistical power, we assume that the estimated treatment effect at endline – a 0.22 percentage point reduction in seed moisture content – represents the true effect. Figure A3 plots a power curve relating sample size to the probability of detecting this effect at the 5% significance level, given the observed baseline variability in moisture content. The curve shows that the null result is unlikely to be driven by the reduced sample size: even with substantially larger samples, an effect of this magnitude would remain undetectable.

We also examine additional indicators of seed performance, including binary variables for whether an agro-dealer received a seed-related customer complaint, a warning following an inspection, or had products confiscated during an inspection. These outcomes are available for larger samples than the indicators derived from purchased seed bags in Table 12. We find no treatment effects on these variables, providing further evidence against detectable improvements in the seed-performance measures available in our data.

Figure A3: Power curve for seed moisture at endline

Table A7: Treatment effects on agro-dealers' seed performance (alternative proxies)

	baseline	*midline*		*endline*	
	mean	treated	obs.	treated	obs.
Shop received seed related complaint from customer†	0.644	-0.089	306	-0.039	297
	(0.480)	(0.057)		(0.048)	
Shop received warning after inspection†	0.317	0.005	291	-0.009	284
	(0.466)	(0.073)		(0.063)	
Shop's products were confiscated after inspection†	0.145	-0.027	293	-0.025	285
	(0.353)	(0.046)		(0.036)	
Overall Anderson index	-0.007	0.066	303	0.060	292
	(0.680)	(0.100)		(0.087)	
Max. number of obs.			306		297

Note: Column (1) reports baseline means and standard deviations below; columns (2) and (4) report differences between treatment and control groups and standard errors below; they are clustered at the level of randomization; columns (3) and (5) report the number of observations; **, *, and + denote significance at the 1%, 5%, and 10% levels; † indicates that the variable is included in the overall index; larger indices indicate more desirable outcomes.